OLYMPIC SPORTS PERFORMANCE ANALYSIS

DOCUMENT 01

Project Overview and Description

Background, objectives, scope, and technical architecture

Academic Project Documentation

Purpose and Scope

This document records the project's approach based on the project context available. Compare all descriptions and counts with the final Tableau workbook and dataset before academic submission.

Data note: Important: project context describes the workbook as using demonstration/synthetic data. Treat findings as findings from this dataset, not verified historical Olympic facts, unless independently verified.

1. Introduction

Olympic Sports Performance Analysis is a data-visualization project designed to explore patterns in sports participation and medal outcomes. It combines a Tableau workbook with a Flask website so viewers can navigate the project's dashboard and story pages in a browser.

2. Problem Statement

Athlete participation, medal totals, sport categories, country performance, and time trends can be difficult to compare when examined separately. This project brings several performance indicators together in a dashboard-based analytical experience.

3. Project Objectives

- Explore medal distribution and performance trends across countries and years.
- Compare participation and medal outcomes using summary metrics.
- Examine differences between regions and sport categories.
- Provide contextual views such as host-country comparisons.
- Present the visual analysis through a browser-based Flask application.

4. Scope

The project context describes coverage from 1992–2026, across 60 countries and 50 sports. The workbook structure includes athlete records, medal results, country-year summaries, investment data, annual medal trends, host-medal reference data, regional efficiency, sport-category metrics, reference metrics, and a data dictionary.

5. Intended Users

The project is intended for students, instructors, and viewers who want to explore sports-performance indicators interactively. It also demonstrates integration of Tableau visualizations into a simple web application.

6. Deliverables

- A Tableau workbook with analytical sheets and dashboard/story views.
- A Flask website with Home, Dashboard, and Story routes.
- A GitHub repository containing the project code.
- A public Render deployment.
- Four documentation PDFs describing the project.

7. Technology Overview

Tableau is used for visual analysis; Flask renders the website pages; HTML templates define page structure; Gunicorn serves the Flask app in production; Render hosts the app; and GitHub stores the source code.

8. Limitations

Conclusions depend on data quality, coverage, and metric definitions. If the dataset is synthetic or illustrative, it cannot establish real-world historical claims. A relationship between investment and medals does not by itself demonstrate causation.